

ABHIPRIYO BISWAS

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PROFESSIONAL SUMMARY

Full Stack Software Engineer with 7+ years of experience designing, building, and shipping production web applications across React.js, Next.js, AngularJS, React Native, and Node.js, including RESTful API design, MongoDB, PostgreSQL/MySQL database schema design, Redux state management, Redis caching, and secure credential management (GCP Secret Manager) for high-traffic enterprise systems. Career progression from Web/UI Designer to Frontend Developer to Full Stack Developer, with end-to-end ownership from Figma-to-code UI implementation through backend API design and production debugging in Agile, cross-functional teams. MSc in Data Science (CGPA 7.78, 2026), adding applied depth in Python ETL pipelines (Airflow, dbt), SQL analytics, and machine learning (Scikit-learn, TensorFlow, PyTorch) — positioned for MERN Stack Developer, Full Stack Developer roles.

TECHNICAL SKILLS

Frontend: React.js, Next.js, AngularJS, Angular, React Native, JavaScript (ES6+), TypeScript, Redux, Redux Toolkit, Redux-Thunk, Zustand, TanStack Query, HTML5, CSS3, Tailwind, Bootstrap, jQuery, Responsive Design, Component-Based Architecture, Storybook

Backend & APIs: Node.js, Express.js, RESTful API Design and Development, Microservices Architecture, Layered API Architecture, JSON, HTTP, Redis (Caching)

Databases & SQL: PostgreSQL, MySQL, SQL, MongoDB, SQLAlchemy, Advanced SQL (Window Functions, CTEs, Joins, Aggregations, Subqueries, Indexing, Query Optimization)

Cloud & DevOps: GCP (BigQuery, Secret Manager, Cloud Logging), Docker, Git, GitHub, Jenkins (CI/CD)

Testing & Quality: Unit Testing, Jest, Enzyme, React Testing Library, Debugging, Production Issue Resolution, Error Handling, Logging, Performance Optimization

AI-Assisted Development: GitHub Copilot, ChatGPT, Claude (Code Review, Planning, Architecture, Productivity)

Web Automation & Scraping: Selenium (JS-Rendered Site Automation), BeautifulSoup, Requests, Fault-Tolerant/Error-Isolated Scraping Architecture, Scheduled Automation (APScheduler/cron)

Data & Analytics (Applied): Python (Pandas, NumPy), SQL Analytics, ETL/ELT Pipelines (Apache Airflow, dbt), Data Visualisation (Plotly, Dash, Power BI), Machine Learning Fundamentals (Scikit-learn, TensorFlow, PyTorch)

Big Data & Orchestration: Apache Airflow, Apache Spark (familiar)

Machine Learning: Scikit-learn, XGBoost, Classification, Regression, Clustering, SMOTE, Cross-Validation, Hyperparameter Tuning, Model Evaluation (ROC-AUC, F1, Precision/Recall), TensorFlow, PyTorch, CNN (Medical Imaging), Transfer Learning

Statistical Analysis: Hypothesis Testing, Descriptive Statistics, Distribution Analysis, Resampling Methods

Tools: VS Code, Jupyter Notebook, Overleaf, Linux

Methodologies: Agile, SDLC, Modular Architecture, End-to-End Project Delivery, Cross-functional & Distributed Team Collaboration, Documentation

PROFESSIONAL EXPERIENCE

Career Progression: Web Designer → Frontend Developer → Full Stack Developer

Consultant – Contractual Employment

Aug 2021 – Dec 2025

TELUS Digital (via Etelligens Technologies)

Remote, Noida, India

- **Engaged as a full-time contractor through Etelligens Technologies.**
- Worked as a Contractor on the **Sapphire team**, contributing to **CASA**, a customer support application used by call center agents to access customer profiles, billing information, service plans, offers, and account dashboards in real time.
- Delivered real-time customer data access for call center agents by developing and enhancing features in a **React.js** and **Node.js – NoSQL data modeling** application, integrating secure backend services to retrieve and display offers data from downstream APIs.
- Shipped production features by implementing **Redux** and **Redux-Thunk** for reliable client-side state management of backend API responses.

- Improved backend maintainability and security by collaborating with API teams to implement a **layered API architecture**, where frontend services consumed intermediary APIs that securely accessed and aggregated data from backend systems.
- Reduced frontend payload complexity by building backend services in Node.js/JavaScript that performed data sanitization, returning only relevant data to the frontend.
- Developed custom, reusable UI components and tested them using **Storybook**, improving component quality and consistency across the application
- Strengthened code reliability by writing unit tests using **React Testing Library** and Jenkins for frontend, and **Jest** for backend; used **Redis** for caching and **GCP Secret Manager** for secure credential storage.
- Maintained live service availability during active customer calls by monitoring **GCP Cloud Logging** for errors and warnings, and debugging/resolving production issues faced by customer care agents in real time.
- Improved system scalability and deployment independence by participating in the organization's transition from monolithic architecture to **microservices**, decomposing application modules into independently deployable services.
- Delivered consistent production stability by working within Agile teams using **Git**-based version control, contributing to production releases across customer-facing systems.
- Worked with **AI** tools **GitHub Copilot**, **ChatGPT** and **Claude** to speed up day-to-day development — scaffolding components, debugging production issues, writing unit tests, and drafting technical documentation.

Front End Developer

Dec 2020 – Dec 2025

Etelligens Technologies

Remote, Noida, India

- Maintained and shipped new features for **TravelNet**, a hybrid mobile application, and **Alloodo**, a game-rental client website, through ongoing frontend maintenance and feature enhancement.
- Developed production-grade **React.js** and **AngularJS** applications, consuming backend REST APIs and optimizing data-driven UI performance across 5+ client projects, including **Pyro**, **Local Kings**, and **Huskarma**.
- Delivered consistent, on-schedule feature releases by collaborating with distributed engineering teams in remote environments across Agile sprint cycles.

Mobile Application Developer

Jun 2020 – Dec 2020

Globiz Technology Inc.

Remote, Ludhiana, India

- Delivered pixel-perfect, cross-browser-tested static websites and **React Native** mobile app screens from design mockups, working across both design and development team responsibilities.
- Delivered performant, production-ready mobile experiences by developing hybrid mobile applications using **React Native**, consuming RESTful APIs and integrating structured backend data, including **LocalCart** and **LineManagement** (deployed to the Google Play Store).

Associate UI Developer

Aug 2018 – May 2020

Promatics Technologies Pvt. Ltd.

Ludhiana, India

- Designed responsive web and mobile application UI using **Photoshop**, **HTML**, **CSS**, and **Bootstrap** across multiple concurrent client projects.
- Supported implementation of GUI designs into functional front-end code alongside the development team.
- **Selected Portfolio:** UCatch, PortaPrep, Babzer, GPS Colombia (Design).

Upskilling & Career Transition

Dec 2025 – Present

Upskilling

Ludhiana, India

- Deepening Backend Development, System Design, and Database expertise following TELUS contract conclusion and completion of MSc Data Science dissertation.
- Shipped 5 independent data engineering/ML projects (below) demonstrating production-grade FastAPI, Airflow/dbt, and CNN-based ML pipelines.

DATA ENGINEERING & MACHINE LEARNING PROJECTS

Job Listings Aggregator — Full-Stack | *React, TypeScript, Redux Toolkit, TanStack Query, Tailwind, Python, Selenium, BeautifulSoup, FastAPI, SQLAlchemy, PostgreSQL, Docker* [Source Code] **2026**

- Built an end-to-end job-search platform: a Python/Selenium/BeautifulSoup scraping pipeline feeding a FastAPI + PostgreSQL backend, consumed by a React/TypeScript dashboard.
- Designed a type-safe API contract between backend (Pydantic) and frontend (TypeScript interfaces) so schema changes fail the build instead of failing silently in production
- Built the frontend with Redux Toolkit (filter/search state) and TanStack Query (server-state caching, background refetching) — deliberately separated, not overlapping responsibilities

- Implemented list virtualization (react-window), debounced search, and route-based code splitting for performance
- Built a **FastAPI** REST API exposing 6+ combinable filters (skill, location, remote status, experience, source, free-text search) and pagination over the aggregated dataset.
- Prevented duplicate postings across repeated scrapes by implementing hash-based deduplication with a database-level unique constraint.
- Automated scraping hourly with deduplication (hash + DB unique constraint) and email digests for new postings
- Shipped CI on both repos (pytest/GitHub Actions on backend, Jest+RTL/GitHub Actions on frontend) — 53 frontend + 11 backend tests, all gating merges
- Ensured reliable, repeatable deployment by containerizing the full pipeline with **Docker** (including headless Chromium + chromedriver); validated correctness with 11 automated tests covering the API, filters, and deduplication logic.

CoinPulse — Crypto Market ETL Platform | *Python, PostgreSQL, Apache Airflow, dbt, Docker, REST API*, **2026**
Data Warehousing [Source Code]

- Eliminated manual data collection and processing by designing and deploying a production-grade end-to-end ETL/ELT pipeline, automating daily extraction and transformation of live cryptocurrency market data (CoinGecko REST API) into a PostgreSQL data warehouse.
- Enforced data quality at the transformation layer by writing **dbt** models converting raw ingested data into structured Silver and Gold layer tables, with built-in dbt tests.
- Prevented duplicate records and maintained data integrity across repeated pipeline executions by implementing idempotent upsert logic and primary-key conflict handling.
- Automated pipeline scheduling and workflow orchestration using **Apache Airflow** DAGs within a Dockerized environment.
- Surfaced historical price trends, asset performance KPIs, and market movement insights across 100+ cryptocurrency assets through SQL analytics reports.
- Improved maintainability and reduced code duplication by designing a modular Python architecture with reusable components and centralized configuration management.
- Ensured reliable ingestion and transformation of API data through comprehensive validation checks and error handling mechanisms.

LogPulse — Application Log Analytics Pipeline | *Python, PostgreSQL, Batch Processing, Data Modeling, SQL*, **2026**
Analytics [Source Code]

- Converted unstructured raw log text into clean, analytics-ready relational data by building a scalable batch data pipeline to ingest and process synthetic application logs into structured records in PostgreSQL.
- Ensured clean, analytics-ready datasets by implementing data parsing, field extraction, and multi-stage data validation checks.
- Enabled monitoring of request volume, HTTP status distribution, and application error trends by developing SQL-based analytical reports and KPI dashboards.
- Supported high-volume log data ingestion with built-in data quality enforcement through scalable, engineered ingestion design patterns.
- Enabled efficient querying and reporting workloads by designing normalized relational schemas and indexing strategies.

Credit Card Fraud Detection — ML Data Pipeline | *Python, Pandas, NumPy, Scikit-learn, XGBoost, Feature Engineering, Data Preprocessing*, **2025**
 [Source Code]

- Produced clean, model-ready data from a transactional dataset containing over 1M+ records by building end-to-end data preprocessing and feature engineering pipelines.
- Improved training data quality under significant class imbalance by applying data cleaning, normalization, and SMOTE-based resampling techniques.
- Delivered standardized, model-ready datasets with validated feature distributions and schema consistency by developing automated transformation workflows compatible with Scikit-learn and XGBoost estimators.
- Derived actionable insights on transaction risk patterns by applying cross-validation and performance benchmarking.

Interactive Movie Analytics Dashboard | *Python, Pandas, NumPy, Plotly, Dash* [Source Code], **2025**

- Enabled non-technical stakeholders to explore movie revenue, ratings, trends, and performance metrics without writing queries, by building a fully interactive, production-ready dashboard.
- Created dynamic visualizations including KPIs, bar charts, trend analysis, and distribution plots, backed by full data preprocessing (cleaning, deduplication, metric creation) and user-driven filters.

- Achieved strong diagnostic classification performance on histopathological image data by developing a deep learning model applying Convolutional Neural Networks (CNNs) with transfer learning to classify malignant vs. benign tissue samples.
- Ensured clinical-grade reliability by conducting statistical analysis of model outputs, including ROC-AUC, precision/recall, and confusion matrix evaluation.
- Improved minority-class recall under class imbalance by applying data augmentation and resampling strategies.
- Demonstrated end-to-end project ownership by producing detailed research documentation covering methodology, experimental design, results analysis, and conclusions.

EDUCATION

Master of Science in Data Science Jul 2024 – Jun 2026

Central University of Haryana, Mahendergarh CGPA: 7.78 — Completed; Degree Certificate Awaited

- Coursework: Data Engineering, Big Data, Database Management Systems (DBMS), Artificial Intelligence, Machine Learning, Statistical Analysis, Data Visualisation.
- Dissertation: Breast Cancer Detection using Deep Learning and Medical Image Classification.

Bachelor of Arts in Computer Science Dec 2020 – Apr 2024

Lovely Professional University

CERTIFICATIONS

HackerRank Skill Certifications: **React (Basic), Angular (Basic), Python (Basic), SQL (Basic), Node (Basic), SQL (Intermediate)**

ADDITIONAL INFORMATION

Languages: English, Hindi, Punjabi, Bengali
Open to: On-site, Remote, Hybrid positions and Contractor engagements
Target Roles: Frontend Developer, Backend Developer, MERN Stack Developer, Full Stack Developer